

National University of Singapore
Semester Examination for the Degree of B.Sc.

(Semester I)

CZ3272 – Monte Carlo and Molecular Dynamics

Nov 2005 – Time Allowed: 2 hours

Instructions to Candidates:

1. This is a closed book examination.
2. This examination paper contains FIVE (5) questions and comprises THREE (3) printed pages.
3. All questions carry equal weight (20 marks each).
4. Answer ALL the questions.

1. Briefly answer the following questions:
 - a. In what condition the probabilities can be added, i.e., $P(A \cup B) = P(A) + P(B)$?
 - b. In what condition the probabilities can be multiplied, i.e., $P(A \cap B) = P(A)P(B)$?
 - c. What is the equation stating the detailed balance in a Markov chain?
 - d. Write down the formula for Metropolis transition probabilities, explain the symbols used.
 - e. How does the Monte Carlo error behave in terms of computer CPU time T .

- a. Probabilities can be added, if the events A and B are mutually exclusive, i.e., $P(A \cap B) = 0$.
- b. Probabilities can be multiplied if event A and B are independent, i.e., $P(A|B) = P(A)$.
- c. $P(X) W(X \rightarrow Y) = P(Y) W(Y \rightarrow X)$
- d. $W(X \rightarrow Y) = T(X \rightarrow Y) \min(1, P(Y)/P(X))$, where $T(X \rightarrow Y) = T(Y \rightarrow X)$ is the probability for choosing a move, and $P(X)$ is equilibrium probability distribution.
- e. $1/\sqrt{T}$.

[4 marks each]

2. In a telephone hotline, it is found that the number of telephones coming in is about 0.10 per minutes, i.e., on average one telephone call every 10 minutes. The distribution of the time interval t between telephone calls is given by the differential equation $dP(t)/dt = -kP(t)$.
 - a. Solve the differential equation and find the normalized probability distribution for the time interval t .
 - b. If the average call is 0.10 per minute, determine the constant k .
 - c. Generate this distribution using a standard, uniformly distributed random number ξ between 0 and 1.

- a. Move $P(t)$ to the left side, and dt to right, and integrate, we get

$$\int dP/P = \int -k dt, \text{ or } \ln P = -kt + c. \text{ That is, } P(t) = P_0 e^{-kt}. \text{ } P_0 \text{ is found by}$$

$$\int_0^{\infty} P_0 e^{-kt} dt = 1. \text{ } P_0 = k.$$
- b. Average t is $\int_0^{\infty} t k e^{-kt} dt = 1/k. = 10 \text{ min, so } k = 0.1 (\text{min}^{-1}).$

c. The exponential distribution $P(t) = k e^{-kt}$ can be generated by inverse method.

First find the distribution function, $F(t) = \int_0^t k e^{-kt'} dt' = 1 - e^{-kt}$. Set $F(t) = \xi$, we

get $t = -(1/k) \ln \xi = -10 \ln \xi$ (min).

[7, 6, 7 marks]

3. Consider a random walker restricted to only three sites, 1, 2, and 3. The walker tries to walk to the left or right with equal probability. If the walker is at the center site 2, it can walk to 1 or 3 with equal probability 1/2; while if it is at the edge 1 or 3, it can walk to 2 with probability 1/2, but the attempted move outside the system is rejected.

- Write down the transition matrix W .
- Is the Markov chain irreducible? Is the Markov chain aperiodic?
- If the walker initially is distributed according to probability $P_1(0)=1/2$, $P_2(0)=1/2$, $P_3(0)=0$, what is the new probability distribution $P_j(2)$, $j=1, 2, 3$, after two steps?
- What is the probability distribution if the walker has executed an infinite number of steps?

a. $W = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{bmatrix}.$

b. Yes, Yes. Because $W^2 = \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 1/4 & 1/2 & 1/4 \\ 1/4 & 1/4 & 1/2 \end{bmatrix}$ is nonzero for all entries. So it

will be nonzero for W^t for $t > 1$.

c. By $P(t) = P(0) W^t$, we find $P(2) = (3/8, 3/8, 1/4)$.

d. Solve the eigenvalue problem, $P_{eq} = P_{eq} W$: Let $P_{eq} = (x, y, z)$, we have, $x = (x+y)/2$, $y = (x+z)/2$, $z = (y+z)/2$, $x+y+z=1$, $x=y=z=1/3$.

[5 marks each]

4. Consider the ratio of two-dimensional integrations:

$$\frac{\int_0^1 dx \int_{-\infty}^{+\infty} dy \log(x+1) e^{-xy^2}}{\int_0^1 dx \int_{-\infty}^{+\infty} dy e^{-xy^2}}.$$

Design a complete pseudo-code of a Metropolis algorithm for the numerical evaluation of the expression.

Take the probability distribution as $P(x, y) = \frac{e^{-xy^2}}{\int_0^1 dx \int_{-\infty}^{+\infty} dy e^{-xy^2}}$, notice that it is

properly normalized. Then the integral is

$$I = \int_0^1 dx \int_{-\infty}^{+\infty} dy P(x, y) \log(x+1) = \langle \log(x+1) \rangle. \quad \text{This is in the standard form}$$

suitable for a Markov Monte Carlo with Metropolis:

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I = 0;
x = 0; y = 0;
for(t = 0; t < T; ++t) {
    x' = x + 0.1 (1-2ξ1);
    y' = y + 0.1(1-2ξ2);
    dE = x'y'2-xy2;
    if(ξ3 < exp(-dE)) {
        x = x';
        y = y';
    }
    I += log(x+1);
}
I = I/T;
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5. In a molecular dynamics simulation of coupled pendulums, one need to simulate the following differential equations:

$$\frac{d^2 x(t)}{dt^2} = -x(t) + x(t)y(t)^2,$$

$$\frac{d^2 y(t)}{dt^2} = -y(t) - x(t)^3 y(t).$$

with the initial conditions $x(0) = 1, y(0)=0, x'(0) = 0, y'(0)=0$.

- Rewrite the equations as a set of first order differential equations, and give the Euler algorithm for a numerical solution of the problem. In particular, specify clearly how the initial conditions will be used in the algorithm.
- Consider solving the same problem with the Verlet algorithm (also known as the Stoeimer algorithm), give the required finite difference equations. Also specify clearly how the initial conditions are used.
- Discuss which of the above two algorithms is better, and why.

- Let $u = dx/dt, v = dy/dt$, then the set of first order differential equations is:

$$dx/dt = u,$$

$$dy/dt = v,$$

$$du/dt = -x + xy^2,$$

$$dv/dt = -y - x^3 y.$$

The Euler algorithm is

$$x_{n+1} = x_n + hu_n,$$

$$y_{n+1} = y_n + hv_n,$$

$$u_{n+1} = u_n - hx_n + hx_n y_n^2,$$

$$v_{n+1} = v_n - hy_n - hx_n^3 y_n.$$

To start, we need set $x_0=1, y_0=0, u_0=0, v_0=0$.

- The Verlet algorithm is

$$x_{n+1} = 2x_n - x_{n-1} - h^2 x_n + h^2 x_n y_n^2,$$

$$y_{n+1} = 2y_n - y_{n-1} - h^2 y_n - h^2 x_n^3 y_n.$$

We can start with $x_0=1, y_0=0, x_1=1-h^2/2, y_1=0$.

- Verlet is better: it is second order, and it is symplectic.

[8, 8, 4 marks]

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